Freefall

Exam code: 0625 & 0972

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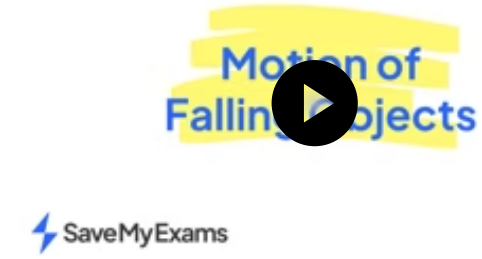

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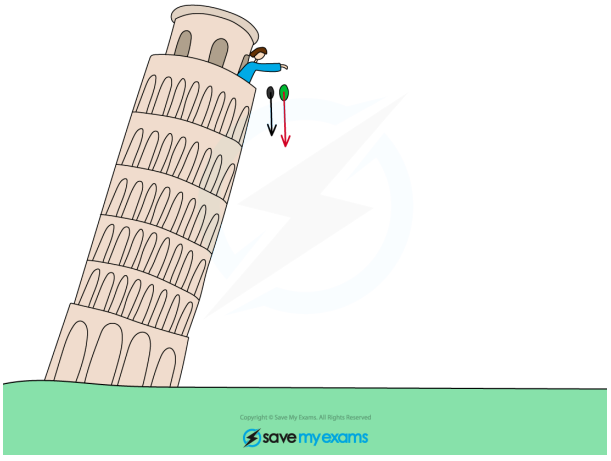
Acceleration of free fall

- In the absence of **air resistance**, all objects fall with the **same acceleration** regardless of their mass
- This is called the **acceleration of freefall**
  - This is also sometimes called acceleration due to gravity

$$\text{acceleration of freefall} = g = 9.8\ \text{m/s}^2$$



In the absence of air resistance, Galileo discovered that all objects (near Earth's surface) fall with an acceleration of

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velocity will increase by 9.8 m/s

- The symbol g also stands for the **gravitational field strength**, and can be used to calculate the force of weight acting an object using its mass:

$$W = mg$$

- Where:
  - W = the force of weight acting on an object, measured in newtons (N)
  - m = mass of object, measured in kilograms (kg)
  - g = gravitational field strength, measured in newtons per kilogram (N/kg)

Motion of falling objects

Extended tier only

Falling objects without air resistance

- A **vacuum** is a space that contains no matter, so there are **no particles** to exert frictional forces on a falling object
- When objects fall in a vacuum, there is no air resistance or liquid resistance so the only force acting on them is the force of **weight**

$$W = mg$$

- Newton's second law of motion describes acceleration as:

$$a = \frac{F}{m}$$

- Where:
  - a = acceleration, measured in metres per second squared (m/s^2)
  - F = force exerted on object, measured in newtons (N)
  - m = mass of object, measured in kilograms (kg)

- Since the only force acting on a falling object in a vacuum is weight, the equation can be expressed as

$$a = \frac{W}{m}$$

- Weight is the product of mass and gravitational field strength, so the equation can be expressed as

$$a = \frac{mg}{m}$$

- Here, the masses cancel each other out

$$a = \frac{\cancel{m}g}{\cancel{m}}$$

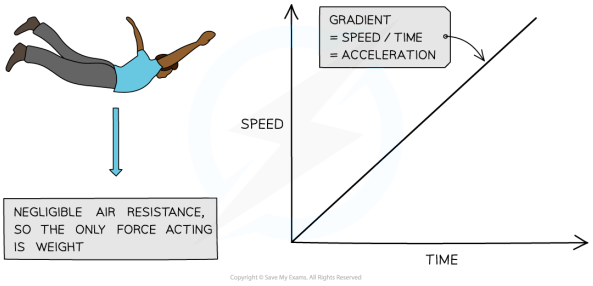
- So, for objects falling in a vacuum

$$a = g$$

- A hammer and a feather were dropped from **equal heights** on the Moon, where there is **no air resistance**
  - The hammer and the feather fell with the **same acceleration** and landed at the same time
  - This proved that objects falling in a vacuum have the same acceleration **regardless of their mass**

- This also applies when air resistance is so small that it can be disregarded
  - When air resistance is described as **negligible**, it can be **approximated** to an object is falling in a vacuum

Object falling with no air resistance



In the absence of air resistance, objects fall with constant acceleration

- Objects falling through a vacuum will **never** reach a **terminal velocity**

Falling objects with air resistance

- When objects fall through a **fluid**, the fluid exerts a **frictional force** on the object as it falls
  - Fluids are **liquids** or **gases**

- Frictional forces **oppose the motion** of an object
  - They act to slow it down

- When an object falls through **air**, it experiences **air resistance**
  - Air resistance is a **frictional** force produced by collisions with air particles as the object moves through the air

- Air resistance **increases** as the **speed** of the object **increases**
- When objects fall through **air**, **two forces** are exerted on the object:
  - The force of **weight**
  - The force of **air resistance**

- When the force of **air resistance** becomes **equal** to the force of weight, then the object **stops accelerating** and falls at a **constant speed**
  - This is called **terminal velocity**

How does a skydiver reach terminal velocity?

The stages of a skydivers fall until they reach terminal velocity

- When a skydiver jumps out of a plane, initially the only force acting on them is **weight**
  - The resultant force acts in the downward direction
  - The skydiver **accelerates**
- As the skydiver accelerates, their **speed increases,** so the force of **air resistance increases**
  - The resultant force acts in the downward direction with a smaller magnitude
  - The skydiver continues to **accelerate** but at a **slower** rate
- **Air resistance** increases until it is **equal** to the **weight**
  - The forces are balanced
  - There is no resultant force
  - **Terminal velocity** is reached
- The parachute is deployed, increasing the surface area of the skydiver
  - The parachute collides with many more air particles
  - **Air resistance increases** greatly
- The force of **air resistance** is now **greater than** the force of **weight**
  - The resultant force acts in the upward direction
  - The skydiver continues falling in a downward direction
  - The skydiver is **decelerating**
- As the skydiver decelerates, their **speed decreases**
  - Therefore, **air resistance decreases**
  - Therefore, the resultant force decreases
- **Air resistance** decreases until it is **equal** to the **weight**
  - The forces are balanced
  - There is no resultant force
  - A new, **slower terminal velocity** is reached

Graph showing the motion of the skydiver as they reach terminal velocity

The graph shows how the speed of the skydiver changes during the descent. The horizontal parts of the graph show the periods of terminal velocity

Worked Example

A small object falls out of an aircraft. Choose words from the list to complete the sentences below:

air resistance **gravitational field strength** **air pressure**

accelerates **falls at a steady speed** **slows down**

(a) The weight of an object is the product of the object's mass and the _____.

(b) When an object falls, initially it _____.

(c) As the object falls faster, the force of _____ acting upon the object increases.

(d) Eventually the object _____ when the force of friction equals the force of weight acting on it.

Answer:

Part (a)

The weight of an object is the product of the object's mass and the **gravitational field strength**.

- The weight force is due to the Earth's gravitational pull on the object's mass as it falls through a uniform gravitational field
-

Part (b)

When an object falls, initially it **accelerates.**

downwards (its weight) - the upward force of air resistance is very small to begin with

Part (c)

As the object falls faster, the force of **air resistance** acting on the object increases.

- The force of **air resistance** is due to **friction** between the object's motion and **collisions with air particles**
- Collisions with air particles slow the object down, so air itself produces a **frictional force**, called **air resistance** (sometimes called **drag**)

Part (d)

Eventually, the object **falls at a steady speed** when the force of friction equals the force of weight acting on it.

- When the upward acting **air resistance** increases enough to **balance** the downward **weight** force, the resultant force on the object is **zero**
- This means the object is no longer accelerating; it is moving at a **steady speed** called **terminal velocity**

Examiner Tips and Tricks

Don't confuse 'air resistance' with 'air pressure' - these are two different concepts!

Exam questions about terminal velocity tend to involve the motion of skydivers as they fall

A common misconception is that skydivers move upwards when their parachutes are deployed; however, this is not the case, they are in fact **decelerating** to a lower terminal velocity

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